University Institute of Information Technology
PMAS-Arid Agriculture University, Rawalpindi

Project Proposal
For

AI-Enabled UGV Hexapod

(18-DOF Dual-Mode Voice-Controlled Robot with Vision Tracking)

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Submission Date (21-09-2026)

Original version

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1. Introduction

The AI-Enabled UGV Hexapod is an 18-DOF six-legged unmanned ground vehicle designed for environments where UAVs and wheeled robots are less suitable. It is intended for confined, GPS-denied, obstructed, and hazardous areas where safe human access is limited or difficult.

The robot combines stable stepping locomotion, embedded control, computer vision, and a web-based control interface to support remote inspection and field monitoring. The mechanical structure consists of six legs with three actuated joints each, allowing the chassis to traverse uneven terrain and narrow passages.

The system supports two primary control methods: a responsive web-based interface and AI-assisted voice interaction through XiaoZhi AI. Users can control movement and predefined actions through either interface, while a pan-tilt FPV camera provides live visual feedback. A lightweight embedded computer-vision model is used for person and object detection/tracking, providing an integrated platform for mobility, remote observation, and human-robot interaction.

2. Literature Review

Indoor UAVs
Indoor UAVs are useful in confined spaces because they can move quickly through the air, but they face challenges in GPS-denied and cluttered environments. They can be affected by obstacles, airflow disturbances, and localization difficulties. Their limited payload capacity and airborne movement can also make close inspection in narrow or obstructed spaces more challenging than ground-based robots.

Wheeled and Tracked UGVs
Wheeled and tracked robots are efficient on relatively flat and continuous terrain, but their mobility can be reduced on uneven surfaces, loose ground, rubble, steps, and gaps. Unlike legged robots, they cannot independently place their limbs to step over obstacles. This can limit their effectiveness in disaster zones, industrial environments, and semi-structured areas where changes in terrain height are common.

Standard Academic Hexapods
Academic hexapods commonly focus on areas such as mechanical design, kinematics, gait generation, and simulation. However, these components are not always developed as part of a single integrated workflow. A robust hexapod platform can combine CAD modelling, robot description, kinematic analysis, gait planning, simulation, and runtime control before physical testing, providing a more structured approach to development and validation.

AI-Based Voice Assistants
AI-based voice assistants provide a natural method for human-robot interaction by allowing users to issue commands through spoken language. XiaoZhi AI provides an ESP32-S3-based voice interaction framework that can connect speech input with AI-assisted conversational and command-based interaction. Its architecture can support local functions such as wake-word detection while computationally intensive speech and language processing may use connected AI services. This makes XiaoZhi suitable for implementing conversational and direct action-based voice control for the proposed hexapod.

3. Benchmarking

Feature	Indoor UAVs	Wheeled UGVs	Academic Hexapod	Proposed UGV
Mobility	Aerial flight	Flat-terrain motion	Legged stepping	18-DOF legged walking
Confined or GPS-denied areas	Limited and unstable	Limited by gaps and obstacles	Good terrain adaptability	Strong stepping mobility
Voice interaction	Usually absent	Usually absent	AI-assisted voice commands	AI-assisted voice commands through XiaoZhi AI
Vision tracking	Optional	Optional	Limited	Face and object tracking
Response time	Variable due to flight control	Moderate & terrain dependent	Low-latency local control	Low local-processing delay
Engineering workflow	Varies	Varies	Often demonstration-focused	CAD, ROS 2, and simulation-first

4. Problem Statement

UAVs and conventional UGVs are limited in constrained and hazardous environments. They may face difficulties in GPS-denied spaces, uneven terrain, debris, and narrow passages, while human operators may be exposed to unnecessary risk when approaching unstable or hazardous structures. In such situations, terrain adaptability and remote observation are important for safe operation.

This project addresses these challenges by designing a stable six-legged UGV that can traverse difficult terrain while keeping the operator at a safe distance. The system combines manual web-based control, AI-assisted voice interaction through XiaoZhi AI, and lightweight computer vision for person and object detection/tracking. The design also integrates mechanical modeling, inverse kinematics, ROS 2 communication, simulation, embedded control, and computer vision into a single engineering workflow.

The proposed platform provides a practical research-oriented system for robotics, embedded systems, computer vision, and human-machine interaction, with applications in inspection, surveillance, emergency response, and other remote-operation scenarios.

A key aspect of the proposed system is the integration of its mechanical, computational, and control components into a unified robotic platform. The six-legged structure is modeled and analyzed using mechanical design tools, while inverse kinematics and gait algorithms are used to generate coordinated leg movements for stable locomotion. ROS 2 provides the communication framework between the robot's control, simulation, and perception modules, allowing the system to be tested and refined before deployment on physical hardware. A web-based human-machine interface enables an operator to remotely control the robot and monitor its operation, while XiaoZhi AI provides voice-based interaction for selected commands. In addition, lightweight computer vision can process the camera feed to assist with basic person and object detection and tracking

5. Problem Solution

The proposed solution is an 18-DOF hexapod robot built to provide stable ground mobility in rough, obstructed, and confined environments. Each leg has three actuated joints, and the system uses inverse kinematics and coordinated gait generation to produce forward movement, turning, strafing, and predefined locomotion behaviors. The robot is designed to step over obstacles and maintain a stable body posture where wheeled systems would lose traction or get stuck.

The voice and vision modules are integrated with the robot to provide responsive human-robot interaction and remote observation. The voice interface supports conversational interaction and direct action commands through XiaoZhi AI, allowing users to control robot movement and predefined behaviors using natural voice commands. The vision module uses a lightweight computer-vision model to perform person and object detection/tracking through the pan-tilt FPV camera.

The camera turret is designed as a modular attachment rather than a fixed-purpose component. Different camera or mission-specific payloads can be mounted according to the intended application, while the core hexapod platform remains unchanged. The development process follows a simulation-driven pipeline from CAD design to URDF/Xacro modeling, ROS 2 control development, Gazebo validation, and final physical prototype testing.

6. Advantages and Benefits of the Proposed System

1. The 18-DOF structure allows coordinated leg movement and better ground contact, making the robot suitable for environments where wheeled robots may have difficulty maintaining stable movement.

2. Enables grounded remote operation in GPS-denied, indoor, narrow, and hazardous environments where UAVs may be unsafe, unstable, or impractical to operate. This allows the robot to approach and observe difficult areas while keeping the human operator at a safer distance.

3. Provides both direct web-based control and AI-assisted voice interaction through XiaoZhi AI. The operator can issue predefined movement and action commands through the control interface or use voice-based interaction for selected robot functions, providing multiple methods of human-robot interaction.

4. Supports remote inspection through a live FPV camera feed, allowing the operator to observe the robot's surroundings in real time. Lightweight computer vision can additionally be used for basic person and object detection and tracking.

5. Uses inverse kinematics and gait planning to calculate and coordinate the movement of all 18 degrees of freedom for walking, turning, and other predefined movements. ROS 2 provides the communication framework between the robot's control, simulation, perception, and other software modules.

6. Follows a structured development workflow using CAD modeling, URDF/Xacro, ROS 2, simulation, circuit validation, and embedded control before full physical testing. The modular design also allows future integration of different cameras, sensors, communication modules, and mission-specific attachments without redesigning the complete robotic system.

7. Scope and Modules

This project covers a six-legged, 18-DOF UGV with two primary user interfaces: a responsive web HCI and a voice AI interface powered by XiaoZhi AI. The web HCI provides robot status and battery information, manual movement control, predefined action control, FPV live video, and turret pan/tilt control. The voice interface provides two modes: Conversational AI mode for natural interaction and Action mode for direct movement and predefined-action commands. The robot can execute forward, backward, left, right, turning, and other locomotion commands using inverse kinematics, gait generation, and coordinated servo control. Predefined actions include Dance, Pushups, Hide, Idle, Slomo, and Run, with the action library designed to remain extensible.

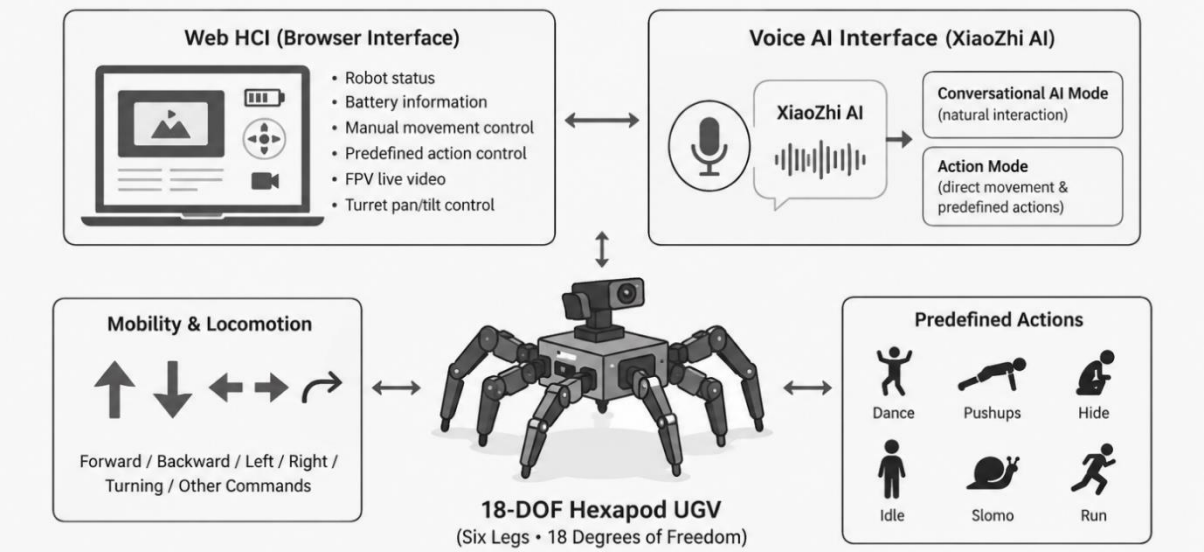


Figure 1: Hexapod System Overview

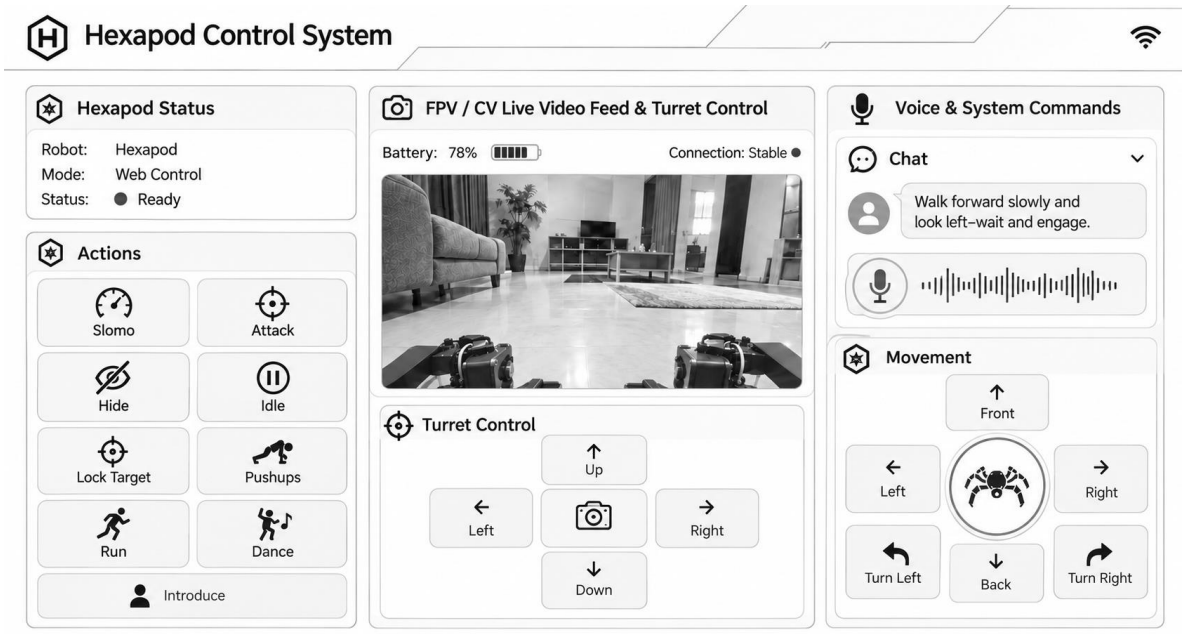


Figure 2: Hexapod Control Interface



Figure 3: UGV Hexapod Physical Prototype

AI and communication layer: Handles voice input/output, command processing, conversational interaction, and action-mode dispatch to the robot control system through the XiaoZhi AI integration.
Robot control layer: Converts motion commands into inverse-kinematics calculations, gait generation, and coordinated leg and servo movement for all 18 DOF.

Motion, vision, and audio layer: Drives locomotion, camera streaming, face and object detection/tracking, and microphone/speaker interaction for real-time operation and remote monitoring.
Physical robot layer: Consists of two ESP32-S3 controllers for system and motion control, one ESP32-CAM for vision and FPV, the hexapod's 18 servo actuators, and a dedicated battery, charging, and power-management system.

Development and validation layer: Includes Fusion 360, ROS 2, URDF/Xacro, RViz, Gazebo, and circuit simulation for design analysis, testing, and iterative improvement before final prototype deployment.

8. Development Methodology and Framework

Stage	Tools	Purpose
1	Mechanical CAD	Design the chassis, legs, and camera mount.
2	URDF / Xacro	Create the robot description and joint limits.
3	ROS 2	Connect control, perception, and locomotion nodes.
4	Git / GitHub	Track versions and support team collaboration.
5	Gazebo / RViz2	Test gaits, joint states, and stability in simulation.
6	Circuit simulation	Check power distribution and connections before assembly.
7	Embedded AI tools	Integrate voice AI and lightweight embedded AI functions.
8	Physical prototype	Integrate and test the complete system.

9. Tools and Technologies

Category	Tool and Technology	Use
Mechanical design	Fusion 360	CAD and mechanical modeling
Robot description	URDF / Xacro	ROS 2 robot model definition
Middleware	ROS 2 Humble	Node communication and control
Simulation	Gazebo	Physics, gait, and stability testing
Visualization	RViz2	Sensor and joint-state monitoring
Circuit checks	Circuit simulation	Power and connection validation
Version control	Git / GitHub	Collaboration and version tracking
Embedded software	ESP32-S3 + ESP32-CAM	Robot control, FPV & embedded vision
Embedded AI	XiaoZhi AI / Embedded AI tools	Conversational AI & action interaction
Computer vision	OpenCV + ESP32-CAM	Face and object tracking
Languages	Python and C++	System integration and robotics logic
Audio	Microphone and speaker	Voice input and output
Power	LiPo battery and power regulation	Logic and actuator power distribution

10. Concepts to be Learned

Inverse Kinematics
Inverse kinematics uses geometric and trigonometric relationships to calculate joint angles that place each leg in the required position. This connects the robot workspace to actual servo commands and is essential for controlling all 18 DOF in a coordinated way.

Robot Description Modeling (URDF and Xacro)
URDF and Xacro describe the robot structure, joints, limits, inertial parameters, and visual geometry. They allow the physical robot to be represented in ROS 2 and tested in simulation before assembly.

ROS 2 Middleware Architecture
ROS 2 provides communication through nodes, topics, services, and actions. It is ideal for connecting locomotion, low-level control, camera processing, and voice interaction into a modular and scalable system.

Physics Simulation and Virtual Validation
Gazebo and RViz2 allow the robot to be evaluated under gravity, inertia, and friction before real-world deployment. This helps detect instability, fine-tune gait strategies, and validate motion plans in a safe and repeatable environment.

Embedded AI / Lightweight Computer Vision
Embedded AI focuses on deploying lightweight models under limited memory, processing, and power constraints. In this project, XiaoZhi AI is used for voice interaction and AI-assisted command processing, while a lightweight computer-vision model is explored for person and object detection/tracking.

Computer Vision and Object Tracking
Computer vision processes video to detect and track faces and objects, with the pan-tilt camera mount used to keep a selected subject centered when required. This capability supports remote monitoring and real-time interaction.

11. Intended Users

Search and Rescue Teams
Emergency responders can use the UGV to inspect unstable disaster areas, collapsed buildings, and narrow pathways without exposing human personnel to high-risk conditions. Multiple FPV cameras can provide different viewing angles for remote inspection and situational awareness.

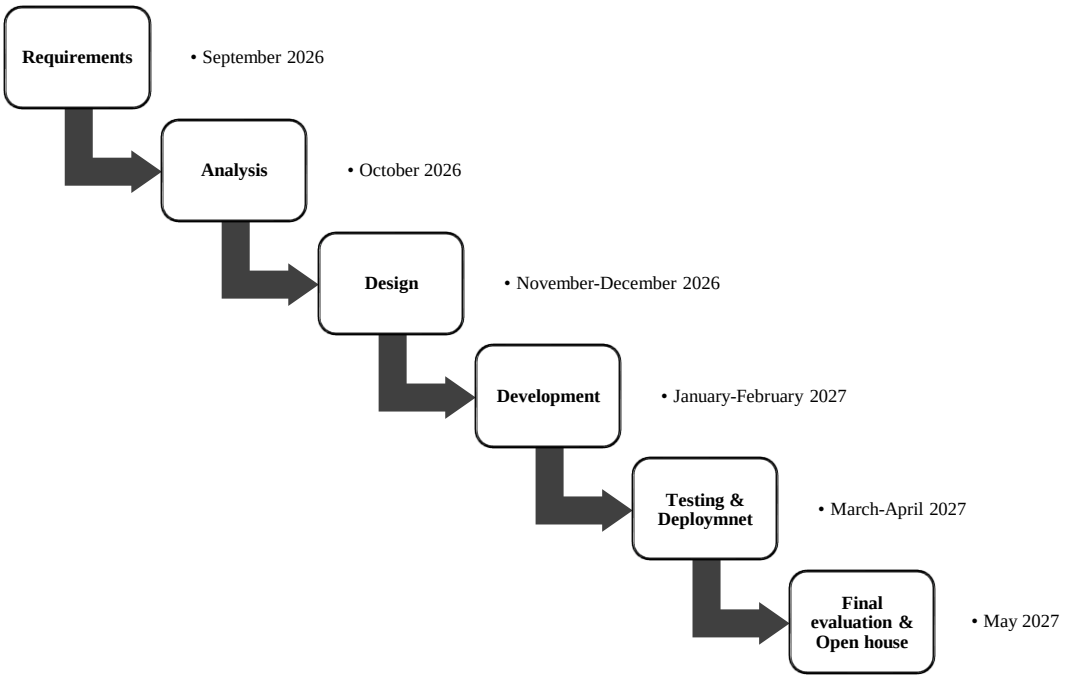
Industrial Inspection Technicians
Inspection teams can use the platform in confined or hazardous facilities, tunnels, plants, and other areas that are difficult for people or wheeled vehicles to access. Different camera modules can be mounted to support visual inspection of multiple areas and angles.

Security and Surveillance Personnel
Security operators can use the robot for remote patrol and observation in restricted or hazardous zones where human presence is undesirable or unsafe. Multiple camera modules can be installed to provide wider visual coverage during remote operation.

Military and Defense Personnel
Military and defense personnel can use the UGV for remote reconnaissance, observation, inspection, and surveillance of potentially hazardous or difficult-to-access environments while reducing direct exposure of personnel to risk. The modular attachment system allows different cameras, sensors, communication equipment, or other mission-specific payloads to be integrated according to operational requirements without changing the core hexapod platform.

Academic and Research Institutions
Universities, laboratories, and robotics researchers can use the platform as an educational and experimental system for studying mobile robotics, inverse kinematics, gait generation, computer vision, embedded systems, ROS 2, and human-robot interaction. The modular platform can also support different camera configurations and additional research modules for future experimentation.

12. Current Project Timeline



13. Conclusion

This project proposes an 18-DOF AI-enabled hexapod UGV designed for stable movement, remote inspection, and operation in confined or hazardous environments. The system combines mechanical design, inverse kinematics, gait control, ROS 2, computer vision, embedded control, and AI-assisted voice interaction.

A responsive web-based HMI will allow the user to control the robot manually, perform predefined actions, view live FPV video, control the camera turret, and monitor basic system status. The system will also support conversational and action-based commands through XiaoZhi AI.

The project provides a practical platform for robotics research and learning. It will help demonstrate concepts such as robot locomotion, inverse kinematics, embedded systems, computer vision, AI integration, and human-machine interaction. The system will be developed in stages, starting with simulation and software testing, followed by hardware integration and real-world testing.

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